Electromagnetic Fields & Waves

Engineering Technology

May, 2026

South East Technological University
Ollscoil Teicneolaíochta an Oirdheiscirt

Electromagnetic Fields & Waves (A15581)

Short Title:	Electromagnetic Fields & Waves		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
Author	PJCREGG		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module covers electromagnetic theory with applications in transmission lines, couplers and antennas. Maxwell's equations are applied to find boundary conditions for static fields. Charge distribution on conducting and insulating blocks in static electric fields are treated with emphasis on the sphere. Solutions for total charge are obtained though integral methods which also serve for radiation pattern analysis. Analysis and design of RF couplers and microstrip lines and assessment of antennas is a key goal. Good mathematical and analytical skills are required for this module.

Programmes

`ELFW-0001`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	3 / 5 / E
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	3 / 5 / M

Indicative Content

- Static Electric Fields: review of Maxwell's eqns, boundary conditions, Laplace equation, field in a trough - analytic solution, numerical approx. and ANSYS.
- Conductors and Insulators: blocks and spheres in fields, induced charges, charge density, charge on partial sphere - by geometric simplification and by analytic integration. a.c quasi-conductors and quasi-insulators.
- Steady Magnetic Fields: review of Maxwell's eqns, boundary conditions, magnetic scalar & vector potentials, force on a dipole.
- Transmission lines and couplers: Impedance of parallel plate, microstrip (Hammerstad approx.), co-axial; RF hybrids - rat race, quadrature hybrid - function & scattering matrix description, phase and group velocity
- Electromagnetic Waves: polarization, power density terms from an antenna: near-field, far-field; 3-d (far field) radiation patterns, beamwidth,directivity, gain, Poynting vector.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Treat the behaviour and characteristics of static electric & magnetic fields at boundaries between media.
2. Derive the capacitance/inductance and characteristic impedance of standard transmission lines.
3. Design basic RF couplers, co-axial cables and transmission lines (co-axial, microstrip).
4. Recognise the basic antennas: short dipole, half-wave dipole and determine the key antenna parameters - beam size, directivity & gain, from radiation patterns.
5. Complete the practical program which involves lab experiments and simulations with reports.

Learning and Teaching Methods

- Lectures

- Tutorials
- Laboratory Work

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	
Practical	20%	5
In-Class Assessment	10%	1,2,3,4

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Practical	12	
Independent Learning	87	

Essential Material

- Guru, B. and H. Hiziroglu. _Electromagnetic Field Theory Fundamentals_. 2nd ed.. Cambridge: Cambridge University Press, 2004.

Supplementary Material

- Balanis, C.A. _Antenna Theory_. 4th ed.. New York: Wiley, 2016.
- Carter, R. _Electromagnetism for Electronic Engineers_. n/a: n/a, 1992.
- Kraus, J. and D. Fleisch. _Elect__romagnetics_. 5th ed.. UK: McGraw Hill, 1999.

- Popovic, Z. and B. Popovic. _Introductory Electromagnetics_. Prentice Hall. .: New Jersey, 2000.

Requested Resources

- ENGINEERING LAB: Electronics